



LIFE LINE TECH - REAL TIME MEDICAL AVAILABILITY TRACKER



A PROJECT REPORT

Submitted by

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*in partial fulfillment of the requirements for the award of
the course*

AMB1331 – DESIGN PROJECT

**DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE
AND MACHINE LEARNING)**

**K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)**

SAMAYAPURAM - 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled “**LIFE LINE TECH - REAL TIME MEDICAL AVAILABILITY TRACKER**” has been carried out by the students **SAMEEHA J, SUDHARSON S R, VETHAABINAYA S, VIMAL AANANTH B R**. The work reported here in is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

Lifeline Tech is a modern web-based application designed to improve emergency healthcare accessibility by enabling users to locate the real-time availability of essential medicines in nearby pharmacies. The system provides a user-friendly interface where individuals can search for medicines, detect their live location using GPS, or manually select their current location from a list of Indian cities or custom input. Based on the selected location, the application dynamically displays nearby pharmacies within a defined radius, along with details such as availability status, distance, and contact information. The platform incorporates an interactive embedded map that accurately marks the real geographic locations of pharmacies using their latitude and longitude, ensuring users can easily identify and navigate to the nearest available medical store. A dedicated “Emergency Mode” feature enhances usability by quickly highlighting the closest pharmacy with the required medicine in stock. In addition to the user module, Lifeline Tech includes a separate admin module for pharmacy owners, allowing them to securely log in and manage their inventory by adding medicines, updating stock levels, and toggling availability status. All updates are reflected in near real-time on the user interface, ensuring data reliability and efficiency. It also minimizes the time required to search for critical medicines during emergencies. The application can be further extended to include features such as notification alerts and alternative medicine suggestions.

Keywords : emergency healthcare, medicine availability, pharmacy locator, real-time tracking, GPS location, nearby pharmacies, emergency mode, interactive map, inventory management, admin module, responsive design

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

In modern healthcare, access to critical medications is essential for the timely treatment of patients. However, patients often face difficulties in locating essential medicines during emergencies due to the unavailability of real-time information about pharmacy stock levels. Traditional methods involve physically visiting multiple pharmacies, which is not only time-consuming but also highly inefficient, especially in urgent medical situations. In such cases, every minute counts. Lifeline Tech aims to bridge this gap by offering a real-time medical availability tracking system that uses location-based services to help users quickly find nearby pharmacies with available stock. The platform utilizes live GPS data, manual location selection, and dynamic search capabilities to deliver accurate information about nearby pharmacies in emergency scenarios. The inclusion of a user-friendly interface ensures that anyone, regardless of their technical expertise, can use the platform effectively to access the required medicines.

1.2 OBJECTIVE

The primary objective of the Lifeline Tech system is to develop a real-time web-based application that enables users to efficiently locate the availability of medicines in nearby pharmacies. The system aims to reduce the time and effort required to search for essential medicines, particularly during emergency situations where immediate access is crucial. Another key objective is to integrate location-based services that allow users to detect their current position using GPS or manually select their desired location, ensuring flexibility and accuracy in search results. The system also focuses on providing a user-friendly interface that simplifies the process of searching for medicines and viewing pharmacy details, including availability status and contact information. Additionally, the project aims to develop a separate admin module for pharmacy owners, allowing them to manage their inventory, update medicine stock, and ensure that the information provided to users is accurate and up to date.

1.3 PROBLEM DEFINITION

The key problem addressed by Lifeline Tech is the lack of a unified system that provides real-time information on the availability of medicines across multiple pharmacies. This gap forces individuals to visit multiple pharmacies or make phone inquiries, resulting in delays, added stress, and unnecessary effort, particularly in emergency situations. Furthermore, existing solutions are often fragmented and lack location-based filtering, making the search for medicines even more difficult.

1.4 OBJECTIVES OF THE STUDY

The primary objective of this project is to develop a web-based platform that offers real-time tracking of medicine availability. By leveraging location-based search and live data updates, the platform aims to reduce the time spent searching for essential medicines during emergencies. The secondary objective is to create an easy-to-use admin interface for pharmacy owners to update their medicine stock, ensuring data accuracy and improving the platform's reliability.

1.5 SCOPE OF THE PROJECT

This project will develop a web application that allows users to search for medicines based on their location. The system will display nearby pharmacies, along with their availability status and contact information. Users will also be able to select a manual location or enable live GPS tracking. The project scope will also include the development of an admin module to allow pharmacies to update their inventory in real time. In the future, the system can be enhanced with features such as mobile apps, online medicine booking, and AI-based medicine suggestions.

CHAPTER 2

LITERATURE SURVEY

2.1 REAL-TIME MEDICINE AVAILABILITY TRACKING SYSTEM USING WEB TECHNOLOGY

AUTHOR : S. Kumar, R. Sharma, P. Gupta

PUBLISHED YEAR : 2023

This study presents a web-based system designed to provide real-time information about medicine availability in nearby pharmacies. The system integrates a centralized database where pharmacy owners can update medicine stock details, allowing users to search and view availability instantly. The application uses location-based services to identify nearby pharmacies and display relevant results. It emphasizes reducing the time required to locate essential medicines during emergencies. The system demonstrates the importance of real-time data synchronization and centralized platforms in improving healthcare accessibility.

MERITS :

- Provides real-time medicine availability information.
- Reduces time spent searching for medicines.
- Uses centralized database for efficient data management.
- Supports location-based pharmacy detection.
- Improves accessibility during emergency situations.

DEMERITS :

- Requires continuous internet connectivity.
- Depends on timely updates from pharmacy owners.
- Limited accuracy if location data is not properly configured.
- Scalability issues with large user data.
- Security concerns in handling sensitive data.

2.2 LOCATION-BASED PHARMACY FINDER USING GPS TECHNOLOGY DETECTOR

AUTHOR : A. Mehta

PUBLISHED YEAR: 2022

This research focuses on a location-based pharmacy finder system that utilizes GPS technology to identify nearby medical stores. The system detects the user's current location and provides a list of pharmacies within a specified radius. It also includes map integration to visually display pharmacy locations, enabling easy navigation. The study highlights the importance of geographic information systems in improving healthcare accessibility. However, the system primarily focuses on location tracking and does not provide real-time medicine availability, which limits its effectiveness during emergencies.

MERITS:

- Uses GPS for accurate location detection.
- Provides map-based visualization of pharmacies.
- Enhances navigation to nearby medical stores.
- Simple and user-friendly interface.
- Improves accessibility to healthcare facilities.

DEMERITS:

- Does not provide real-time medicine availability.
- Accuracy depends on GPS signal strength.
- Limited functionality beyond location tracking.
- No inventory management system.
- Cannot prioritize emergency requirements.

2.3 ONLINE PHARMACY MANAGEMENT SYSTEM WITH INVENTORY TRACKING

AUTHOR: R. VERMA, D. CHAUHAN, S. IYER

PUBLISHED YEAR: 2023

This study introduces an online pharmacy management system that enables pharmacy owners to manage their inventory digitally. The system allows administrators to add, update, and monitor medicine stock levels through a web-based interface. It improves efficiency in managing pharmacy operations and reduces manual errors. Although the system supports inventory tracking, it is primarily designed for internal management and does not provide a direct interface for users to search medicine availability in real time. The research emphasizes the importance of digital inventory systems in modern healthcare environments.

MERITS:

- Efficient inventory management system.
- Reduces manual errors in stock handling.
- Provides centralized data storage.
- Improves operational efficiency of pharmacies.
- Easy to update and manage medicine records.

DEMERITS:

- Not user-oriented for public access.
- Lacks real-time search functionality for users.
- Limited integration with location-based services.
- Requires training for pharmacy staff.
- No emergency-focused features.

2.4 MAP-BASED HEALTHCARE SERVICE FINDER USING WEB APPLICATIONS

AUTHOR : P. NAIR, S. KRISHNAN, M. JOSEPH

PUBLISHED YEAR: 2022

This research presents a web application that integrates mapping technologies to help users locate healthcare services such as hospitals and pharmacies. The system uses interactive maps to display locations and allows users to view details such as address and contact information. The study highlights the importance of spatial visualization in improving user experience. However, the system does not include real-time data updates or medicine availability tracking, which limits its usefulness in emergency healthcare scenarios.

MERITS:

- Provides interactive map visualization.
- Enhances user navigation experience.
- Displays location-based healthcare services.
- Easy to use and visually intuitive.
- Supports multiple healthcare facility searches.

DEMERITS:

- No real-time data updates.
- Does not track medicine availability.
- Limited to location display only.
- No admin control for updating data.
- Not suitable for emergency decision-making.

2.5 REAL-TIME DATA SYNCHRONIZATION IN HEALTHCARE SYSTEMS USING CLOUD TECHNOLOGY

AUTHOR: K. REDDY, M. SHAH, L. FERNANDES

PUBLISHED YEAR: 2024

This study explores the use of cloud-based systems for real-time data synchronization in healthcare applications. The system enables instant updates of medical data across multiple users and devices, ensuring data consistency and reliability. It highlights the importance of real-time communication between users and service providers. The research demonstrates how cloud technology can improve healthcare efficiency by providing up-to-date information. However, the system requires strong network connectivity and proper data security measures to ensure safe operation.

MERITS:

- Supports real-time data synchronization.
- Ensures data consistency across systems.
- Improves efficiency of healthcare services.
- Enables scalable system architecture.
- Supports multi-user access.

DEMERITS:

- Depends on stable internet connection.
- Security risks in cloud-based storage.
- Requires proper data management strategies.
- High implementation cost.
- Needs regular maintenance and monitoring.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In the existing healthcare scenario, users rely on conventional methods to locate medicines, which primarily involve visiting nearby pharmacies physically or contacting them individually to inquire about availability. This process is often time-consuming and inefficient, especially during emergency situations where immediate access to medicines is critical. In most cases, there is no centralized system that provides real-time information about medicine availability across multiple pharmacies. As a result, users are forced to visit several medical stores before finding the required medicine, leading to delays and inconvenience. Some digital platforms and online pharmacy applications exist, but they are mainly focused on medicine ordering and do not provide accurate real-time stock information for nearby pharmacies. Additionally, these systems often depend on fixed location data and do not allow users to modify their location dynamically, resulting in irrelevant or inaccurate search results. The absence of integrated map-based visualization further limits the ability of users to identify the exact location of pharmacies. Due to the lack of real-time updates and proper coordination between pharmacies and users, the existing system fails to meet the requirements of emergency healthcare situations effectively.

3.1.1 DEMERITS:

- Detection of medicine availability relies on manual search, leading to delays and inefficiency.
- Users often need to visit multiple pharmacies to find a specific medicine.
- Lack of real-time data results in outdated or inaccurate information.
- No centralized system connecting pharmacies and users.
- Limited or no support for dynamic location selection.
- Absence of integrated map visualization for easy navigation.

3.2 PROPOSED SYSTEM

The proposed system, Lifeline Tech, introduces a web-based real-time medical availability tracking solution designed to overcome the limitations of existing methods. The system enables users to search for specific medicines and instantly retrieve information about nearby pharmacies that have the required stock. It integrates advanced location-based services that allow users to detect their current location using GPS or manually select and modify their location according to their preference. This ensures accurate and flexible search results in different scenarios.

The system incorporates an interactive map that displays the real geographic locations of pharmacies using latitude and longitude coordinates. Each pharmacy is represented as a marker on the map, allowing users to visually identify nearby medical stores and access detailed information such as availability status and contact number. Unlike traditional systems, the proposed solution provides real-time or near real-time updates, ensuring that users receive accurate and up-to-date information. The system also includes an emergency mode feature that prioritizes the nearest pharmacy with the required medicine, enabling quick decision-making during critical situations.

3.2.1 MERITS:

- Enables real-time medicine availability tracking.
- Reduces time required to locate essential medicines.
- Supports both live GPS and manual location selection.
- Provides interactive map with accurate pharmacy locations.
- Displays contact information for easy communication.
- Includes emergency mode for faster decision-making.
- Allows admin to update inventory in real time.

CHAPTER 4

SYSTEM SPECIFICATION

4.1 SOFTWARE SPECIFICATION

- Operating System : Windows 10 / Windows 11 / Android / iOS
- Programming Language : JavaScript (Frontend), Node.js / Python (Backend)
- Framework : React.js / Express.js (or equivalent)
- Database : MongoDB / Firebase
- Mapping Library : Leaflet / OpenStreetMap
- Web Browser : Google Chrome / Microsoft Edge / Mozilla Firefox

4.1.2 FRONTEND SOFTWARE

- HTML, CSS, JavaScript
- Bootstrap / Tailwind CSS
- React.js (for dynamic user interface)
- Leaflet.js (for interactive map visualization)
- Axios / Fetch API (for API communication)

4.1.3 BACKEND SOFTWARE

- Node.js / Express.js
- REST API for data handling
- Authentication (JWT / Session-based login)
- MongoDB / Firebase for database management

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The proposed system follows a unified client–server architecture designed to support real-time medical availability tracking based on user location. The architecture integrates modern web technologies with real-time data management systems to provide fast, accurate, and user-friendly access to pharmacy information. The system is divided into two main components: the Client Side and the Server Side, which communicate through a backend API to ensure seamless data flow and synchronization.

On the client side, the system provides a web-based user interface developed using HTML, CSS, and JavaScript, along with modern frontend frameworks for enhanced responsiveness. The interface allows users to search for medicines, view availability status, and interact with an embedded map. A location service module is integrated using browser-based Geolocation API, which enables the system to capture the user’s live GPS location. Additionally, the system provides a manual location selection feature that allows users to change their location by selecting a city or entering custom input. This ensures flexibility and accurate results even when GPS detection is unavailable or incorrect.

The client interface communicates with the backend through RESTful APIs, sending user requests such as medicine search queries and location data. These requests are processed by the backend API, which is implemented using Node.js or Flask. The backend acts as a central communication layer that manages data processing, request handling, and interaction between different system components. It receives user inputs, processes search queries, and retrieves relevant pharmacy and medicine data from the database.

On the server side, the system utilizes a centralized database such as MongoDB or Firebase to store pharmacy details, including name, address, geographic coordinates, contact number, and medicine inventory. Each pharmacy record is associated with its latitude and longitude, enabling accurate mapping and distance calculation. When a user searches for a medicine, the backend performs location-based filtering to identify nearby pharmacies within a specified radius and checks the availability status of the requested medicine.

The system also integrates an interactive mapping engine using technologies such as Leaflet and OpenStreetMap. This component is responsible for rendering the map on the client interface and displaying pharmacy locations as markers based on their geographic coordinates. Each marker provides additional details, including pharmacy name, availability status, and contact number when selected by the user. The map dynamically updates whenever the user changes their location, ensuring that results always correspond to the selected area.

To ensure real-time or near real-time updates, the system incorporates a synchronization mechanism that reflects changes made in the admin module instantly on the user interface. The admin module allows pharmacy owners to update medicine stock, which is then stored in the database and immediately accessible to users. The backend ensures efficient handling of concurrent requests and maintains consistency between data updates and user queries.

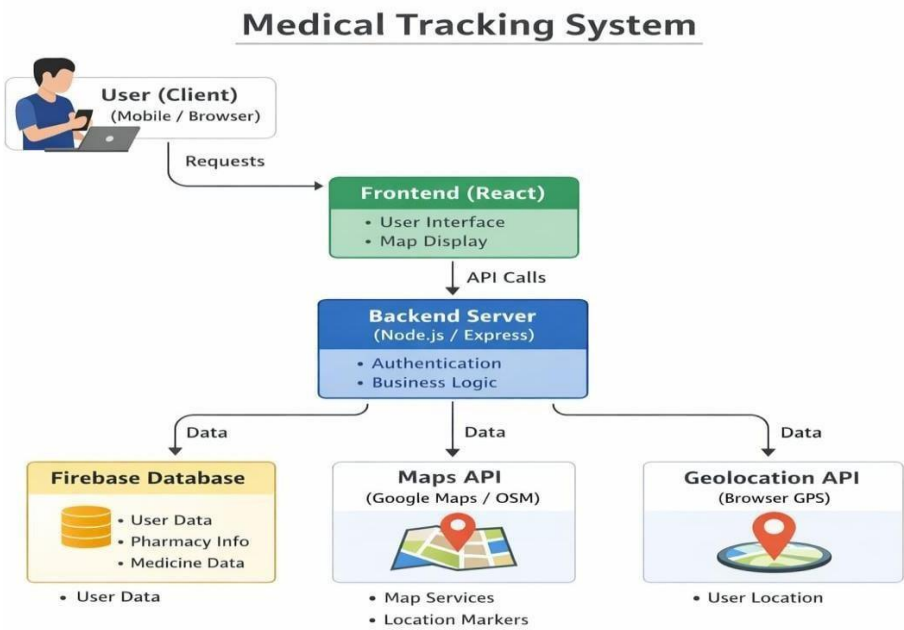


Fig 5.1 System Architecture

Overall, the system architecture is designed to be scalable, efficient, and responsive, enabling seamless interaction between users and pharmacy services. By combining location-based services, real-time data management, and interactive visualization, the Lifeline Tech system provides a reliable and effective solution for medical availability tracking.

CHAPTER 6

MODULES DESCRIPTION

6.1 METHODOLOGY

6.1.1 ALGORITHM IMPLEMENTATION

The main algorithm of the Lifeline Tech system begins with collecting the user's input, which includes the name of the required medicine and the user's location. The location can be obtained either through the browser-based GPS geolocation service or through manual selection of a city or custom input provided by the user. The input data is validated and converted into a standardized format to ensure consistency in processing.

Next, the system performs location-based filtering by calculating the distance between the user's location and the stored coordinates of registered pharmacies. The database contains detailed information about pharmacies, including their latitude, longitude, and available medicines. Using this data, the system identifies pharmacies within a predefined radius and checks whether the requested medicine is available in stock.

The filtered results are then processed and sorted based on proximity and availability status. Pharmacies with the required medicine in stock are prioritized and displayed first. The system also integrates a mapping engine that plots the pharmacy locations using geographic coordinates and displays them as markers on an interactive map.

Finally, the processed data is sent back to the user interface, where it is presented in both list and map formats. Each result includes details such as pharmacy name, distance, availability status, and contact number. The system ensures smooth communication between all components and handles errors such as invalid input, unavailable location data, or missing medicine records.

6.1.2 USER INPUT AND LOCATION MODULE

The User Input and Location Module is the initial stage of the Lifeline Tech system, where the user interacts with the application by entering the required medicine and selecting their location. The system provides flexibility by allowing users to either use live GPS-based location detection or manually select their current location from a list of cities or by entering custom input. This ensures that users are not restricted by inaccurate GPS readings and can search based on their preferred area.

The module captures and validates user input to ensure correctness and completeness before processing. It also handles scenarios where location access is denied or unavailable by providing alternative input options. The use of browser-based geolocation services enables accurate positioning, while manual selection enhances usability and accessibility. This module plays a critical role in ensuring that the system receives accurate input, which directly affects the quality of search results.

6.1.3 DATA PREPROCESSING MODULE

The Data Preprocessing Module is responsible for preparing the input data for efficient processing. Once the user provides the medicine name and location, the system standardizes the data by converting it into a consistent format. This includes normalizing text inputs, removing unnecessary characters, and validating location coordinates.

The module also processes location data by converting addresses or city names into geographic coordinates using geocoding techniques. This ensures compatibility with distance calculation and mapping functions. Additionally, the system filters out invalid or incomplete data entries to prevent errors during processing. By ensuring that all input data is clean and standardized, the preprocessing module improves the accuracy and reliability of the system's output.

6.1.4 DATABASE AND DATA MANAGEMENT MODULE

The Database and Data Management Module serves as the central repository for storing all system-related information. It maintains detailed records of pharmacies, including their name, address, geographic coordinates, contact number, and available medicines along with stock status. The database is designed to support efficient data retrieval and real-time updates.

When a user initiates a search, the system queries the database to retrieve relevant pharmacy data based on location and medicine availability. The module ensures fast data access and supports concurrent user requests. It also maintains data consistency by synchronizing updates made by pharmacy administrators. This centralized data management approach enables the system to provide accurate and up-to-date information to users.

6.1.5 LOCATION PROCESSING AND FILTERING MODULE

The Location Processing and Filtering Module is responsible for identifying nearby pharmacies based on the user's location. The system calculates the distance between the user's coordinates and each pharmacy's location using geographic distance formulas. Only pharmacies within a specified radius are considered for further processing.

The module then filters these pharmacies based on the availability of the requested medicine. Pharmacies that have the medicine in stock are prioritized, while others may be marked as unavailable. This filtering process ensures that users receive relevant and meaningful results. The module also supports dynamic updates, allowing the results to change instantly when the user modifies their location. By combining location-based filtering with availability checks, the system provides accurate and efficient search results.

6.1.6 MAP VISUALIZATION MODULE

The Map Visualization Module is responsible for displaying the geographic locations of pharmacies on an interactive map. The system uses mapping technologies such as Leaflet and OpenStreetMap to render the map within the application. Each pharmacy is represented as a marker placed according to its latitude and longitude coordinates.

Users can interact with these markers to view additional details such as pharmacy name, availability status, and contact number. The map dynamically updates based on the user's selected location, ensuring that only relevant pharmacies are displayed. This visual representation enhances user understanding and makes it easier to locate nearby medical stores. The integration of map-based visualization significantly improves the usability and effectiveness of the system.

6.1.7 ADMIN MODULE

The Admin Module provides a secure interface for pharmacy owners to manage their inventory and update medicine availability. Administrators can log in to the system and perform operations such as adding new medicines, updating stock levels, and modifying pharmacy details. These updates are stored in the database and reflected in real time on the user interface.

The module includes validation mechanisms to ensure that only authorized users can access and modify data. It also supports efficient data entry and management, reducing manual errors. By enabling real-time updates, the admin module ensures that users receive accurate and up-to-date information about medicine availability. This module plays a crucial role in maintaining the reliability and effectiveness of the system.

6.1.8 RESULT DISPLAY AND OUTPUT MODULE

The Result Display and Output Module is the final stage of the system, where processed data is presented to the user in a clear and user-friendly format. The system displays a list of nearby pharmacies along with details such as medicine availability, distance, and contact number. The results are also shown on an interactive map, providing a visual representation of pharmacy locations.

The module ensures that information is displayed in an organized and easily understandable manner, enabling users to make quick decisions. It also handles cases where no results are found by providing appropriate messages. The integration of both textual and visual output enhances user experience and ensures effective communication of information. This module completes the system workflow by delivering accurate and actionable results to the user.

CHAPTER 7

RESULT AND DISCUSSION

7.1 COMPARATIVE STUDY

The comparative analysis illustrates the performance differences between the existing system and the proposed Lifeline Tech system across key parameters such as accuracy, efficiency, real-time capability, scalability, and user convenience. The results clearly indicate that the proposed system significantly outperforms the existing system in all aspects. In terms of accuracy, the proposed system provides more reliable results by offering real-time medicine availability information directly from pharmacies, reducing the chances of incorrect or outdated data. Similarly, the system improves efficiency by minimizing the time required for users to locate essential medicines, thereby enhancing the overall user experience.

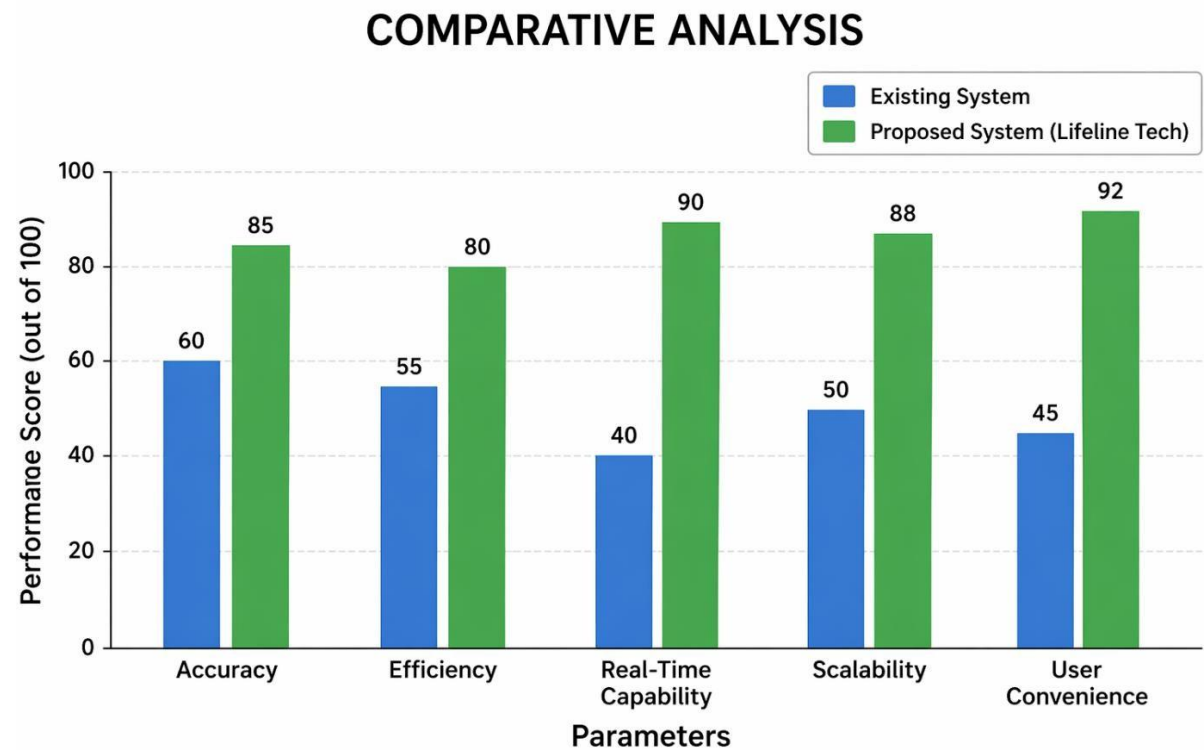


Figure 7.1 Comparative Analysis

The proposed system demonstrates considerable improvement in real-time capability and scalability, making it highly suitable for practical implementation in modern healthcare environments. By integrating live location tracking and dynamic data updates, the system enables users to instantly access information about nearby pharmacies and medicine availability. This feature is particularly beneficial in emergency situations where quick decision-making is essential. In contrast, the existing system relies heavily on manual searching and static data, which limits its ability to provide immediate and accurate results.

In terms of efficiency, the proposed system reduces the need for physical visits to multiple pharmacies by providing a centralized platform for medicine search. This not only saves time but also reduces user effort and inconvenience. The system also shows improved scalability, as it can handle multiple users and large volumes of data simultaneously without affecting performance. This makes it suitable for deployment in both urban and rural areas.

Furthermore, the proposed system enhances user convenience by incorporating features such as interactive map visualization, live location detection, and manual location selection. These features provide flexibility and improve accessibility for users. The inclusion of an admin module ensures that pharmacy data is regularly updated, which further increases the reliability of the system. In comparison, the existing system lacks these advanced features, making it less effective in meeting the demands of real-time healthcare services.

Overall, the comparative study clearly demonstrates that the Lifeline Tech system is more efficient, accurate, scalable, and user-friendly than traditional methods. By combining real-time data processing, location-based services, and interactive visualization, the proposed system provides a practical and advanced solution for medical availability tracking, thereby improving healthcare accessibility and response time.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

The Lifeline Tech system demonstrates the effective use of modern web technologies and location-based services in addressing one of the critical challenges in healthcare, which is the timely availability of essential medicines. By integrating real-time data management, geolocation services, and interactive map visualization, the system provides a reliable platform for users to quickly locate medicines in nearby pharmacies. The structured workflow, including user input handling, location processing, database management, and result visualization, ensures accurate and efficient functioning of the system.

The implementation of features such as live location detection, manual location selection, and dynamic data filtering enables the system to deliver precise and personalized results. The use of an interactive mapping interface further enhances user experience by visually representing pharmacy locations and enabling easy navigation. Additionally, the system minimizes the time and effort required for users to search for medicines, thereby improving accessibility and convenience, especially during emergency situations.

The inclusion of a dedicated admin module ensures that pharmacy data is continuously updated, maintaining the accuracy and reliability of the system. Real-time synchronization between the admin and user modules allows instant reflection of inventory changes, which is essential for maintaining trust and usability. Compared to traditional methods, the proposed system significantly improves efficiency, reduces delays, and enhances decision-making.

Overall, the Lifeline Tech system provides a scalable, efficient, and user-friendly solution for real-time medical availability tracking. It contributes to improved healthcare accessibility by ensuring that essential medicines can be located quickly and reliably, thereby supporting timely medical intervention and better patient outcomes.

8.2 FUTURE ENHANCEMENT

The Lifeline Tech system provides a strong foundation for real-time medical availability tracking; however, several enhancements can further improve its functionality and effectiveness. One important area of improvement is the expansion of the database by integrating a larger number of pharmacies across different regions. This would increase system coverage and ensure that users can access information from a wider network of medical stores. Incorporating more detailed inventory data, including alternative medicines and generic substitutes, can also enhance the usefulness of the system.

Another significant enhancement involves the integration of advanced technologies such as artificial intelligence and predictive analytics. AI-based recommendation systems can suggest alternative medicines when the required medicine is unavailable, improving user satisfaction. Predictive analytics can be used to analyze demand patterns and forecast medicine availability, helping pharmacies manage stock more efficiently. Additionally, incorporating voice-based search functionality can make the system more accessible, especially for elderly users and individuals with limited technical knowledge.

Further improvements can include the integration of online reservation or booking features, allowing users to reserve medicines before visiting the pharmacy. Real-time notification systems can also be implemented to alert users when a medicine becomes available in nearby locations. The system can be extended to support multilingual interfaces, making it more accessible to users from different linguistic backgrounds.

Deployment enhancements such as cloud-based infrastructure and mobile application development can improve scalability and accessibility. Integration with hospital systems and emergency services can further enhance the system's impact by providing a more comprehensive healthcare solution. Additionally, implementing advanced security measures and data privacy protocols will ensure safe handling of user and pharmacy data. These enhancements will make the Lifeline Tech system more robust, scalable, and adaptable to real-world healthcare environments.

APPENDIX A

SOURCE CODE

```

<!doctype html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <link rel="icon" type="image/svg+xml" href="/vite.svg" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>LifeLine Tech</title>

  </head>

  <body>

    <div id="root"></div>

    <script type="module" src="/src/main.tsx"></script>

  </body>

</html>

```

```

import { BrowserRouter as Router, Routes, Route, Link, Navigate, useNavigate }
from 'react-router-dom';

import { AppProvider, useAppContext } from '../context/AppContext';

import { Stethoscope, LogOut } from 'lucide-react';

import Home from './pages/user/Home';

import SearchResults from './pages/user/SearchResults';

import AuthPage from './pages/Auth';

import AdminDashboard from './pages/admin/AdminDashboard';

```



```
const RequireUser = ({ children }: { children: JSX.Element }) => {
```

```

const { user, authLoading } = useAppContext();

if (authLoading) return <div style={{ padding: '4rem', textAlign: 'center' }}>Authenticating securely...</div>;

return user ? children : <Navigate to="/auth" />;

};

const RequireAdmin = ({ children }: { children: JSX.Element }) => {

  const { isAdmin } = useAppContext();

  return isAdmin ? children : <Navigate to="/auth" />;

};

const Header = () => {

  const { user, isAdmin, logout } = useAppContext();

  const navigate = useNavigate();

  const handleLogout = async () => {

    await logout();

    navigate('/auth');

  };

  return (

    <header style={{

      background: 'rgba(255, 255, 255, 0.7)',

      backdropFilter: 'blur(16px)',

      WebkitBackdropFilter: 'blur(16px)',

      borderBottom: '1px solid rgba(255, 255, 255, 0.5)',

      padding: '1.5rem 0',

      position: 'sticky',

      top: 0,

```

zIndex: 50,

boxShadow: 'var(--shadow-sm)'

}}>

<div className="container" style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center' }}>

<Link to="/" style={{ display: 'flex', alignItems: 'center', gap: '0.75rem', textDecoration: 'none', color: 'var(--primary)', fontWeight: 800, fontSize: '1.75rem', letterSpacing: '-0.5px' }}>

<Stethoscope size={36} strokeWidth={2.5} />

LifeLine Tech

</Link>

<nav style={{ display: 'flex', gap: '2rem', alignItems: 'center' }}>

{(user || isAdmin) && (

<

{user && <Link to="/" style={{ textDecoration: 'none', color: 'var(--text-main)', fontWeight: 600, fontSize: '1.1rem', transition: 'color 0.2s' }}
onMouseOver={(e) => e.currentTarget.style.color = 'var(--primary)'}
onMouseOut={(e) => e.currentTarget.style.color = 'var(--text-main)'}>Find
Medicine</Link>}

{isAdmin && <Link to="/admin" style={{ textDecoration: 'none', color: 'var(--text-main)', fontWeight: 600, fontSize: '1.1rem', transition: 'color 0.2s' }}
onMouseOver={(e) => e.currentTarget.style.color = 'var(--primary)'}
onMouseOut={(e) => e.currentTarget.style.color = 'var(--text-main)'}>Admin
Dashboard</Link>}

<button

onClick={handleLogout}

className="btn btn-secondary"

style={{ height: 'auto', padding: '0.6rem 1.25rem', fontSize: '0.95rem', gap: '0.5rem', borderRadius: 'var(--radius-full)' }}

>

```

        <Logout size={16} /> Logout

      </button>

    </>

  )}

</nav>

</div>

</header>

);

};

function App() {

  return (

    <AppProvider>

      <Router>

        <div style={{ display: 'flex', flexDirection: 'column', minHeight: '100vh' }}>

          <Header />

          <main style={{ flex: 1 }}>

            <Routes>

              <Route path="/auth" element={<AuthPage />} />

              <Route path="/" element={<RequireUser><Home /></RequireUser>} />

              <Route path="/results" element={<RequireUser><SearchResults
/></RequireUser>} />

              <Route path="/admin" element={<RequireAdmin><AdminDashboard
/></RequireAdmin>} />

              <Route path="*" element={<Navigate to="/" />} />

            </Routes>

```

```

    </main>

    </div>

  </Router>

</AuthProvider>

);
}

export default App;

import React, { useState } from 'react';

import { useNavigate } from 'react-router-dom';

import { auth } from '../firebase';

import { signInWithEmailAndPassword, createUserWithEmailAndPassword } from
'firebase/auth';

import { Lock, User, ShieldAlert, Mail } from 'lucide-react';

import { useAppContext } from '../context/AppContext';

export default function AuthPage() {

  const [isLogin, setIsLogin] = useState(true);

  const [isAdminTab, setIsAdminTab] = useState(false);

  const [email, setEmail] = useState("");

  const [password, setPassword] = useState("");

  const [adminUser, setAdminUser] = useState("");

  const [error, setError] = useState("");

  const [loading, setLoading] = useState(false);

```

```

const navigate = useNavigate();

const { loginAsAdmin, loginMockUser } = useAppContext();

const handleUserAuth = async (e: React.FormEvent) => {
  e.preventDefault();

  setError("");

  setLoading(true);

  try {
    // Auto-fallback simulation if Firebase configuration isn't loaded yet!
    if (auth.app.options.apiKey?.includes('MOCK')) {
      console.warn("Using strict Firebase MOCK mode. Bypassing real network authentication.");

      loginMockUser(email);

      navigate('/');

      return;
    }

    if (isLogin) {
      await signInWithEmailAndPassword(auth, email, password);

      navigate('/');
    } else {
      await createUserWithEmailAndPassword(auth, email, password);

      navigate('/');
    }
  } catch (err: any) {
    console.error(err);
  }
}

```

```

    if (err.code === 'auth/invalid-credential' || err.code === 'auth/wrong-password' ||
err.code === 'auth/user-not-found') {

        setError('Invalid username or password');

    } else if (err.code === 'auth/api-key-not-valid') {

        setError('Missing Configuration: Please add your actual Firebase API keys to
the `.env` file to process real signups.');
```

} else {

 setError(err.message || 'Authentication failed');

 }

 } finally {

 setLoading(false);

 }

};

const handleAdminAuth = (e: React.FormEvent) => {

 e.preventDefault();

 if (adminUser === 'ADMIN' && password === '2026') {

 loginAsAdmin();

 navigate('/admin');

 } else {

 setError('Invalid Admin Credentials');

 }

};

return (

 <div style={{ padding: '4rem 1.5rem', display: 'flex', justifyContent: 'center',
minHeight: 'calc(100vh - 80px)' }}>

```
<div className="card" style={{ width: '100%', maxWidth: '450px', padding: '2.5rem' }}>
```

```
  <div style={{ display: 'flex', marginBottom: '2rem', borderBottom: '2px solid var(--border)' }}>
```

```
    <button
```

```
      style={{ flex: 1, padding: '1rem', background: 'transparent', border: 'none',
borderBottom: !isAdminTab ? '3px solid var(--primary)' : '3px solid transparent',
fontWeight: !isAdminTab ? 700 : 500, fontSize: '1.1rem', color: !isAdminTab ? 'var(--primary)' : 'var(--text-muted)', cursor: 'pointer', transition: 'all 0.2s' }}
```

```
      onClick={() => { setIsAdminTab(false); setError(""); }}
```

```
    >
```

```
      User Portal
```

```
  </button>
```

```
  <button
```

```
    style={{ flex: 1, padding: '1rem', background: 'transparent', border: 'none',
borderBottom: isAdminTab ? '3px solid var(--primary)' : '3px solid transparent',
fontWeight: isAdminTab ? 700 : 500, fontSize: '1.1rem', color: isAdminTab ? 'var(--primary)' : 'var(--text-muted)', cursor: 'pointer', transition: 'all 0.2s' }}
```

```
    onClick={() => { setIsAdminTab(true); setError(""); }}
```

```
  >
```

```
    Admin Portal
```

```
</button>
```

```
</div>
```

```
<h2 style={{ textAlign: 'center', color: 'var(--text-main)', marginBottom: '1.5rem',
fontSize: '1.75rem', fontWeight: 700 }}>
```

```
  {isAdminTab ? 'Secure Admin Login' : (isLoggedIn ? 'Welcome Back' : 'Create Account')}
```

```
</h2>
```



```

{error && (
  <div style={{ background: 'var(--danger-bg)', color: 'var(--danger)', padding:
'1rem', borderRadius: 'var(--radius-md)', marginBottom: '1.5rem', display: 'flex',
alignItems: 'center', gap: '0.5rem', fontWeight: 500 }}>

    <ShieldAlert size={20} />

    {error}

  </div>

)}

<form onSubmit={isAdminTab ? handleAdminAuth : handleUserAuth}
style={{ display: 'flex', flexDirection: 'column', gap: '1.25rem' }}>

  <div className="input-group">

    {isAdminTab ? (

      <User style={{ position: 'absolute', left: '1rem', top: '50%', transform:
'translateY(-50%)', color: 'var(--text-muted)' }} size={20} />

      ) : (

        <Mail style={{ position: 'absolute', left: '1rem', top: '50%', transform:
'translateY(-50%)', color: 'var(--text-muted)' }} size={20} />

        )}

    <input

      type={isAdminTab ? "text" : "email"}

      placeholder={isAdminTab ? "Admin Username" : "Email Address"}

      style={{ paddingLeft: '3rem' }}

      value={isAdminTab ? adminUser : email}

      onChange={e => isAdminTab ? setAdminUser(e.target.value) :
setEmail(e.target.value)}

```

```

        required

    />

</div>

<div className="input-group">

    <Lock style={{ position: 'absolute', left: '1rem', top: '50%', transform:
'translateY(-50%)', color: 'var(--text-muted)' }} size={20} />

    <input

        type="password"

        placeholder="Password"

        style={{ paddingLeft: '3rem' }}

        value={password}

        onChange={e => setPassword(e.target.value)}

        required

    />

</div>

<button type="submit" className="btn btn-primary" style={{ marginTop:
'1rem', width: '100%', height: '3.5rem' }} disabled={loading}>

    {loading ? 'Processing...' : (isAdminTab ? 'Access Dashboard' : (isLoggedIn ?
'Login to LifeLine Tech' : 'Sign Up securely'))}

</button>

</form>

{!isAdminTab && (

    <div style={{ textAlign: 'center', marginTop: '2rem', color: 'var(--text-muted)'
}}>

        {isLoggedIn ? "Don't have an account?" : "Already have an account?"}

    <button

```

```

    type="button"

    onClick={() => setIsLogin(!isLogin)}

    style={{ background: 'none', border: 'none', color: 'var(--primary)',
fontWeight: 600, fontSize: '1rem', cursor: 'pointer', marginLeft: '0.5rem' }}

    >

    {isLogin ? 'Sign Up' : 'Log In'}

  </button>

</div>

  )}

</div>

</div>

);

```

APPENDIX B

SCREENSHOTS

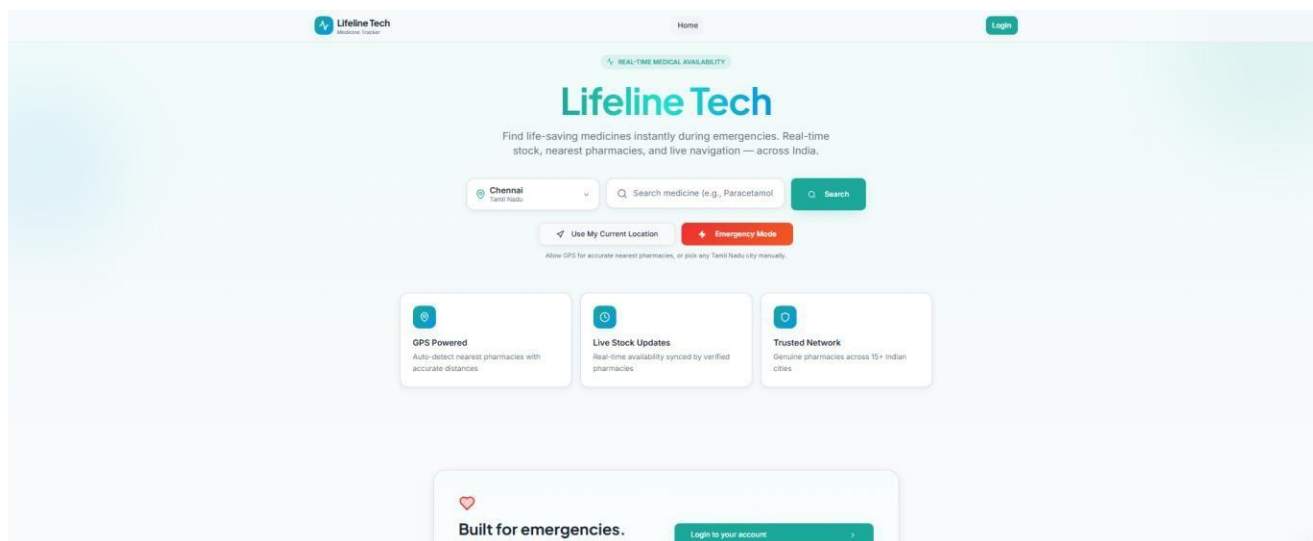


FIGURE.B.1 HOMEPAGE PERFORMANCE RESULT

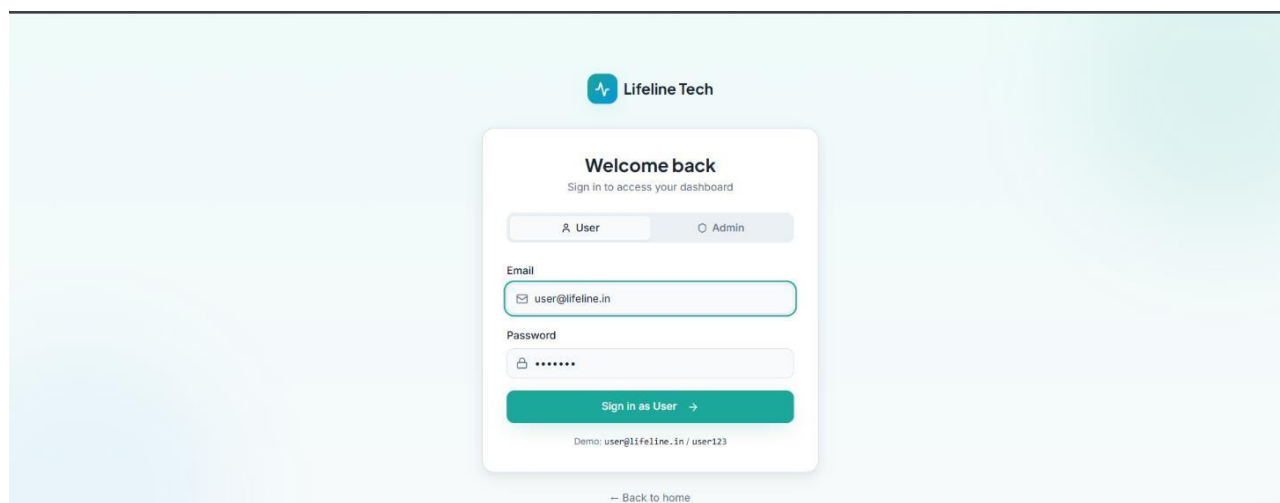


FIGURE.B.2 USER AUTHENTICATION RESULT

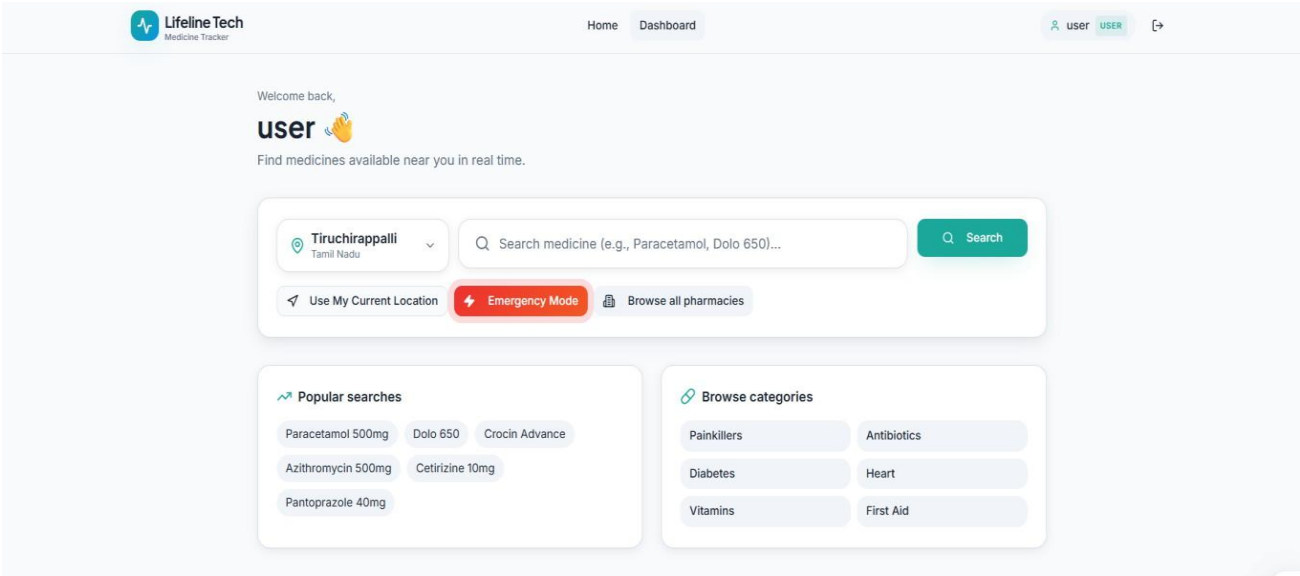


FIGURE.B.3 DASHBOARD FUNCTIONALITY RESULT

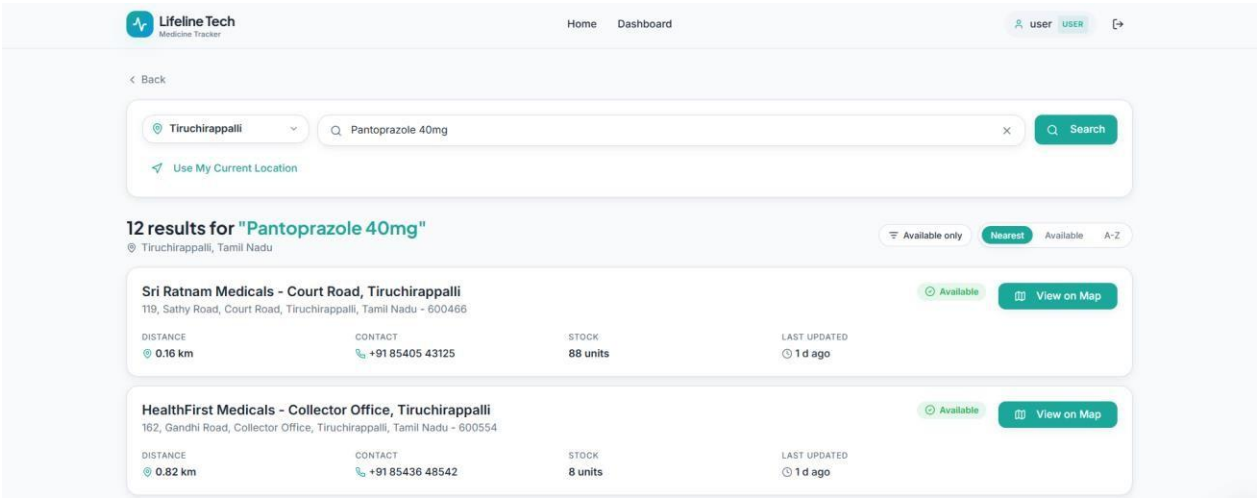


FIGURE.B.4 MEDICINE SEARCH RESULT

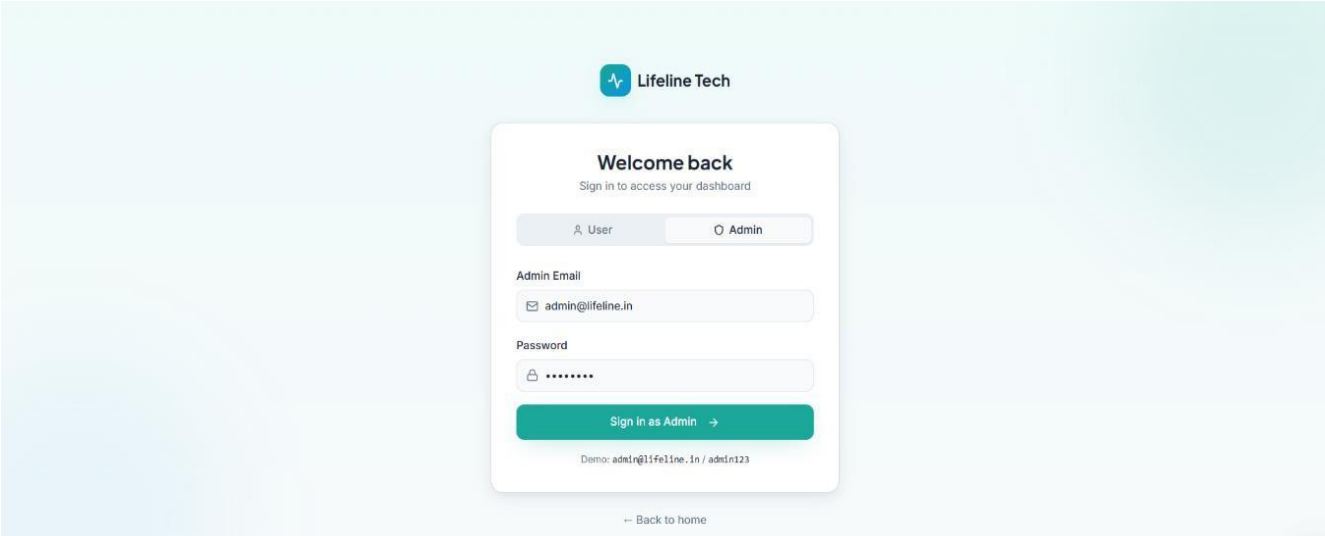


FIGURE.B.5 ADMIN AUTHENTICATION RESULT

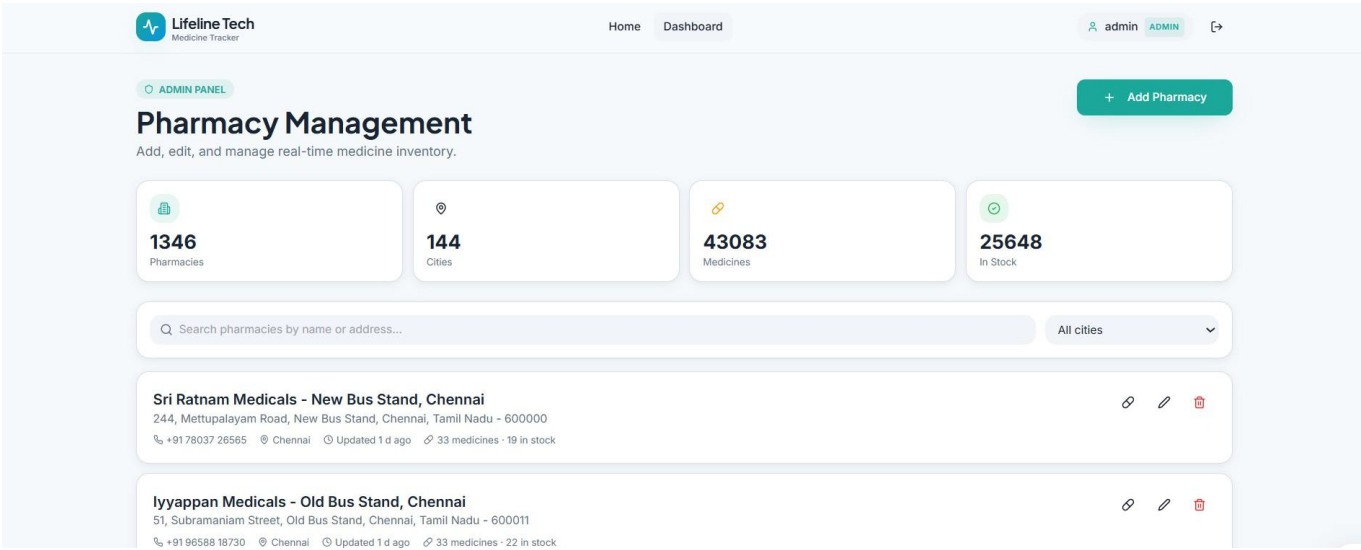


FIGURE.B.6 PHARMACY MANAGEMENT MODULE

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